

Heart Health Analysis - Solution Guide

Basic Level Questions (10)

Q1. Gender Distribution of Patients

Answer:

The dataset contains both male and female patients. Male patients represent a larger proportion of the dataset, indicating a higher representation of males in the study.

Business Insight:

Understanding gender distribution helps identify population groups requiring targeted cardiovascular screening.

Business Impact:

Healthcare providers can design gender-specific awareness and prevention programs.

Q2. Average Age of Patients

Answer:

The average patient age was calculated using the age column.

Business Insight:

Most patients belong to middle-aged and older populations where cardiovascular risks are higher.

Business Impact:

Age-based screening programs can improve early detection.

Q3. Distribution of Chest Pain Types

Answer:

Typical Angina was the most commonly observed chest pain category.

Business Insight:

Chest pain characteristics are important indicators of cardiovascular conditions.

Business Impact:

Early identification of high-risk chest pain symptoms can improve diagnosis accuracy.

Q4. Average Resting Blood Pressure

Answer:

The average resting blood pressure provides an overview of cardiovascular health among patients.

Business Insight:

Elevated blood pressure is a major risk factor for heart disease.

Business Impact:

Regular monitoring can reduce complications and healthcare costs.

Q5. Average Maximum Heart Rate

Answer:

The maximum heart rate achieved during exercise was analyzed for all patients.

Business Insight:

Heart rate response can indicate cardiovascular fitness and disease risk.

Business Impact:

Exercise recommendations can be personalized based on heart rate performance.

Q6. Heart Disease Distribution

Answer:

The dataset includes both heart disease and non-heart disease patients, allowing comparative analysis.

Business Insight:

Understanding prevalence helps evaluate population risk.

Business Impact:

Supports healthcare planning and resource allocation.

Q7. Average Cholesterol Level

Answer:

Average Cholesterol Level = **246.50 mg/dL**

Business Insight:

Many patients exhibit cholesterol levels above the ideal range.

Business Impact:

Cholesterol management programs can reduce future cardiovascular complications.

Q8. Resting ECG Distribution

Answer:

- ST-T Abnormality: 151
- Normal: 147
- LVH: 4

Business Insight:

Abnormal ECG findings are common among patients.

Business Impact:

Routine ECG screening can improve early diagnosis.

Q9. Patients with Resting ECG Result = 2

Answer:

Only **4 patients** had a resting ECG result of 2 (LVH).

Business Insight:

LVH cases are relatively rare in the dataset.

Business Impact:

Early intervention can prevent further cardiovascular complications.

Q10. Distribution of Major Vessels (ca)

CA Patients

0	175
1	65
2	38
3	20
4	4

Business Insight:

Most patients have zero major vessels identified through fluoroscopy.

Business Impact:

Vessel count is an important indicator of disease severity.

Mid-Level Questions (10)

Q1. Correlation Between Age and Cholesterol

Answer:

Correlation = **0.207**

Insight:

Weak positive relationship exists between age and cholesterol.

Business Impact:

Older patients may benefit from more frequent cholesterol monitoring.

Q2. Chest Pain Distribution Across Age Groups

Answer:

The 50–60 age group contained the highest number of patients across all chest pain categories.

Business Insight:

Middle-aged patients exhibit the highest cardiovascular burden.

Business Impact:

Age-specific treatment and awareness programs can improve outcomes.

Q3. Maximum Heart Rate vs Exercise-Induced Angina

Condition Avg Heart Rate

No Angina 155.60

Angina 137.21

Insight:

Patients with exercise-induced angina show lower maximum heart rates.

Business Impact:

Exercise programs should be customized for higher-risk patients.

Q4. Resting Blood Pressure by Gender

Gender Avg BP

Female 133.08

Male 130.91

T-Test P-Value: 0.318

Insight:

No statistically significant difference exists.

Business Impact:

Similar blood pressure management strategies may apply to both genders.

Q5. Fasting Blood Sugar vs Heart Disease

FBS	No Heart Disease	Heart Disease
≤120 mg/dL	116	141
>120 mg/dL	22	23

Insight:

Heart disease occurs in both blood sugar groups.

Business Impact:

Blood sugar should be considered alongside other risk factors.

Q6. Major Vessels vs Heart Disease

CA	No Disease	Heart Disease
0	45	130
1	44	21
2	31	7
3	17	3
4	1	3

Insight:

Patients with CA=0 show the highest heart disease prevalence.

Business Impact:

Vessel analysis improves risk assessment.

Q7. Average Oldpeak by Chest Pain Type

Chest Pain Type Oldpeak

Typical Angina	1.38
Atypical Angina	0.32
Non-Anginal Pain	0.81
Asymptomatic	1.39

Insight:
Asymptomatic and Typical Angina patients exhibit higher ST depression.

Business Impact:
Supports severity assessment and treatment prioritization.

Q8. Thalassemia Distribution vs Heart Disease

Thal	No Disease	Heart Disease
Unknown	1	1
Fixed Defect	12	6
Normal	36	129
Reversible Defect	89	28

Insight:
Normal thalassemia category contains the largest number of heart disease patients.

Q9. Common Risk Factor Combinations

Answer:
Several chest pain, fasting blood sugar, exercise angina, and thalassemia combinations appear frequently among heart disease patients.

Business Impact:
Supports targeted intervention strategies.

Q10. Clinical Measurement Comparison

Metric	No Heart Disease	Heart Disease
Age	56.60	52.58
BP	134.39	129.25
Cholesterol	251.09	242.64

Metric	No Heart Disease	Heart Disease
Max Heart Rate	139.10	158.38
Oldpeak	1.59	0.59

Insight:

Maximum heart rate shows the largest difference.

Advanced Questions (5)

Q1. Combined Effect of Age, Cholesterol and Blood Pressure

Answer:

Heart disease patients generally exhibited lower average age, cholesterol, and blood pressure compared to non-heart disease patients.

Business Impact:

Multiple variables should be evaluated together rather than individually.

Q2. Strongest Correlation with Heart Disease

Feature	Correlation
Chest Pain Type	0.432
Thalach	0.420
Slope	0.344
Exang	-0.436
Oldpeak	-0.429
CA	-0.409

Strongest Positive Predictor: Chest Pain Type

Strongest Negative Predictor: Exercise-Induced Angina

Q3. Logistic Regression Analysis

Accuracy: 81.97%

Top Positive Predictors

- Chest Pain Type
- Slope
- Resting ECG

Top Negative Predictors

- Sex
- Exercise-Induced Angina
- Thalassemia
- Major Vessels

Business Impact:

Predictive models can support early diagnosis and preventive care.

Q4. Slope Variation by Chest Pain Type

Insight:

Flat and Downsloping patterns dominate across most chest pain categories.

Business Impact:

ECG patterns can improve diagnostic accuracy.

Q5. Thalassemia and Heart Disease Across Age Groups

Insight:

Heart disease prevalence varies across thalassemia types and age groups.

Business Impact:

High-risk thalassemia groups may require additional cardiovascular monitoring.

Final Conclusion

The analysis identified **Chest Pain Type, Maximum Heart Rate, Exercise-Induced Angina, Major Vessel Count, Thalassemia, and Oldpeak** as the most influential factors associated with heart disease. The Logistic Regression model achieved **81.97% accuracy**, demonstrating strong predictive capability for identifying patients at risk of heart disease.